Neuroscience and Biobehavioral Reviews 187 (2026) 106723



Contents lists available at ScienceDirect

Neuroscience and Biobehavioral Reviews

journal homepage: www.elsevier.com/locate/neubiorev





Systematic review of virtual reality therapy in the treatment of visual dysfunction in children with ADHD and dyslexia

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ARTICLE INFO

Keywords:
Visual Dysfunction
Attention deficit hyperactivity disorder
Dyslexia
Binocular dysfunction
Virtual Reality
Virtual Reality Therapy

ABSTRACT

Attention deficit hyperactivity disorder (ADHD) and dyslexia are prevalent neurodevelopmental disorders that significantly impact educational and social functioning. Both conditions are frequently associated with comorbid visual dysfunction, which can further exacerbate learning and attentional difficulties. Virtual reality therapy (VRT) has emerged as a promising intervention for addressing visual impairments such as convergence insufficiency and binocular dysfunction in affected children. This systematic review evaluates the effectiveness of VRT in treating visual dysfunction in children with ADHD and/or dyslexia. 20 studies published between 2015 and 2025 were analyzed. VRT improved visual acuity (0.89–1.32 lines) and stereoacuity in amblyopia, reduced omission/commission errors (14.75/4.75 errors) in ADHD, and enhanced reading speed ($r = 0.992$) and visual perception ($d = 0.93$) in dyslexia, with gains comparable to patching for binocular dysfunction. VRT is an effective intervention for improving visual acuity, binocular function, visual attention, and processing speed in children with ADHD and/or dyslexia, with effect sizes ranging from moderate to large. Key strengths include its engaging, gamified design, which improves treatment adherence. However, the literature remains fragmented: most studies examine ADHD or dyslexia in isolation, with limited research on their comorbidity. Heterogeneity in study designs also constrains definitive conclusions. Future research should prioritize longitudinal studies with large, well-designed randomized controlled trials with longitudinal follow-up to evaluate sustained efficacy.

1. Introduction

Attention deficit hyperactivity disorder (ADHD) is a highly prevalent neurodevelopmental disorder, affecting 5–8% of children and adolescents worldwide (Ayano et al., 2023; Polanczyk et al., 2014; Thomas et al., 2015). This disorder is characterized by symptoms including inattention, hyperactivity, and impulsivity, which can hinder a child's ability to focus on tasks and control their impulses, causing educational and social challenges. ADHD can be classified into three subtypes: predominantly inattentive, predominantly hyperactive-impulsive, and combined (Núñez-Jaramillo et al., 2021). Dyslexia is a learning disability characterized by difficulties in decoding (reading), writing, and spelling (Snowling et al., 2020). Children with dyslexia have impaired phonological abilities, leading to difficulties in decoding words, reading fluently, and comprehending written text. While ADHD and dyslexia are distinct conditions, they frequently occur together, with a high, bidirectional comorbidity rate of 25–40% (McGrath and Stoodley, 2019). This high rate of co-occurrence underscores the need for transdiagnostic interventions that can simultaneously address the overlapping visual and cognitive challenges in children with ADHD and dyslexia.

Importantly, both ADHD and dyslexia are associated with visual dysfunction, a broad term encompassing impairments in the efficient processing of visual information. In the present review, "visual dysfunction" and "visual processing deficits" are defined consistently with the neurodevelopmental literature (Kibby et al., 2015; Raghuram et al., 2019; Stein, 2019). This construct includes (1) low-level visual functions such as amblyopia (reduced visual acuity), binocular

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https://doi.org/10.1016/j.neubiorev.2026.106723
Received 16 January 2026; Received in revised form 16 April 2026; Accepted 29 April 2026
Available online 3 May 2026
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dysfunction (e.g., convergence insufficiency), stereopsis deficits, and accommodative disorders, as well as (2) higher-order visual processing domains, including visual attention and vigilance, visual processing speed, visuospatial working memory, visual perceptual skills (e.g., discrimination, figure-ground perception), and resistance to visual crowding. These higher-order domains are not general cognitive functions but modality-specific visual-cognitive processes that are frequently impaired in ADHD and dyslexia and directly contribute to academic difficulties such as poor reading fluency, inefficient visual search, and sustained attention problems during visual tasks (Cunha et al., 2023; Facoetti et al., 2001; Kibby et al., 2015).

The prevalence of these visual issues is substantial. For instance, 79% of children with dyslexia had a clinically diagnosed deficit in at least one domain of visual function, including accommodation and ocular motor tracking (Raghuram et al., 2018), while children with ADHD show significant impairments in convergence function, including reduced near point of convergence (NPC), compared to their typically developing peers (Bellato et al., 2023). These visual dysfunctions are not merely co-occurring conditions but can play a significant role in the clinical profile. For children with dyslexia, deficits in the magnocellular visual system can lead to unstable binocular control, poor fixation, and visual crowding, directly affecting reading fluency and accuracy while contributing to visual stress, discomfort, and reading avoidance (Stein, 2019). In ADHD, impairments in oculomotor control (Lee et al., 2023) and processing speed (Lee, 2023) have been documented, and such deficits may exacerbate core inattention symptoms, hindering the ability to follow instructions or complete visual tasks efficiently.

Current treatments for visual dysfunction include vision therapy and colored lenses, which offer benefits but often struggle with compliance, engagement, and variable efficacy, especially in pediatric populations (Smaakjær et al., 2022; Suttle and Conway, 2025). Virtual reality therapy (VRT) has emerged as a promising alternative, utilizing computer-generated three-dimensional simulations to create immersive environments for users. This technology has found applications across various fields, including rehabilitation and mental health (Freeman et al., 2024; Lau et al., 2025; Tang et al., 2025).

Recent research suggests that VRT effectively treats visual dysfunctions, improving binocular vision and alleviating symptoms associated with conditions like convergence insufficiency and accommodative dysfunction (Li et al., 2022). Additionally, it holds potential for addressing visual impairments, such as binocular dysfunction, stereopsis, myopia, and strabismus (Backus et al., 2018; Cepeda-Zapata et al., 2019; Fortenbacher et al., 2018; Xu et al., 2024; Yaramothu et al., 2019). In the context of neurodevelopmental disorders, including ADHD and dyslexia, VRT also offers innovative avenues for intervention. For children with ADHD, VRT extends beyond merely minimizing distractions. It enables targeted cognitive training through gamified tasks delivered in immersive virtual classrooms or real-world simulations. This approach has been shown to improve sustained attention, working memory, and executive function by providing controlled, engaging environments where users can practice self-regulation strategies without real-world consequences (Bashiri et al., 2017; Corrigan et al., 2023), though the existing evidence is based on studies with methodological limitations and small sample sizes (Corrigan et al., 2023). Recent meta-analytic evidence confirms that VR-based interventions produce a moderate reduction in attention deficit symptoms (SMD = −0.33), with immersive VR showing particular promise for enhancing ecological validity in assessment and training (Zheng et al., 2025). Similarly, for children with dyslexia, VRT offers multisensory, embodied learning experiences that target both phonological and visual-spatial processing deficits. By using immersive, interactive exercises, VR reduces performance anxiety, facilitates the visualization of texts, and fosters greater motivation, while also allowing for customizable, intensive practice that can be tailored to individual needs (Maresca et al., 2022). Emerging evidence indicates that VR-based neurorehabilitation can produce significant gains in reading fluency and processing speed that are sustained over time (Maresca et al., 2024). Thus, VRT presents innovative avenues for treatment, potentially increasing the effectiveness of interventions for children facing these challenges.

Previous systematic reviews have examined VR interventions either for isolated visual conditions (e.g., amblyopia or strabismus) or for broad cognitive outcomes in ADHD/dyslexia, without specifically synthesizing evidence on the full spectrum of visual processing deficits—including both low-level binocular/oculomotor functions and higher-order visual-cognitive domains—in children with ADHD and/or dyslexia. Most studies have investigated VRT either for general (primarily low-level) visual dysfunctions or for the core symptoms of ADHD and dyslexia independently, leaving a critical synthesis gap at the intersection of these areas.

To bridge this gap, this systematic review aims to consolidate the current state of knowledge, evaluate the effectiveness of VRT in treating visual dysfunctions in children with ADHD and/or dyslexia, and identify methodological limitations and underexplored areas in the literature. This review not only provides an updated synthesis of evidence but also highlights directions for future research to advance the clinical application of VRT in this complex patient group.

Specifically, this research focuses on the following research questions:

RQ1. What visual processing domains are targeted by virtual reality therapy (VRT) interventions in children with ADHD and/or dyslexia?

RQ2. What VRT methodologies and outcome measures are currently employed in treating visual dysfunction in children?

RQ3. What is the effectiveness of VRT in treating visual dysfunctions in children with comorbid ADHD and/or dyslexia?

2. Literature review

2.1. Visual dysfunction in ADHD

Beyond the core symptoms of inattention, hyperactivity, and impulsivity, children with ADHD frequently experience co-occurring visual dysfunction that may exacerbate academic and attentional challenges. A growing body of evidence indicates that ADHD is associated with impairments in multiple visual domains, including convergence (e.g., reduced near point of convergence), accommodation, and visual processing speed (Alqahtani et al., 2019; Bellato et al., 2023; DeCarlo et al., 2016). These visual deficits are well-documented in children with ADHD and are not merely secondary symptoms. A recent systematic review and meta-analysis confirmed a significant association between ADHD and vision problems, including convergence insufficiency and accommodative dysfunction (Bellato et al., 2023).

Importantly, conventional ADHD treatments—including medication, cognitive training, and behavioral therapies—rarely target these visual deficits directly (Jones et al., 2020; Marx et al., 2015; Pan et al., 2019). This oversight is significant because impairments in visual attention, processing speed, and oculomotor control can directly hinder a child's ability to follow visual instructions, complete worksheets efficiently, and sustain focus on academic tasks (Cunha et al., 2023; Kim et al., 2024). Consequently, addressing visual dysfunction may represent an underexplored but potentially valuable avenue for improving functional outcomes in children with ADHD, which is where VRT offers promise through its ability to deliver engaging, adaptive, and ecologically valid visual training.

2.2. Visual dysfunction in dyslexia

Dyslexia, characterized by persistent difficulties in reading, spelling, and decoding, is fundamentally linked to phonological processing deficits (Snowling et al., 2020). Primary interventions, such as phonological-based training and transcranial direct current stimulation

(tDCS), aim to address these core linguistic weaknesses (Costanzo et al., 2016; Pasqualotto and Venuti, 2020). However, the role of visual dysfunction in dyslexia has been a persistent and evolving area of inquiry. Several influential theoretical frameworks have been proposed. The magnocellular theory posits a deficit in the visual pathway responsible for timing, motion perception, and binocular control, which may lead to unstable visual perception and reading difficulties (Stein, 2019). The visual attention span deficit hypothesis argues for a reduced capacity to process multiple visual elements simultaneously, affecting orthographic processing (Bosse et al., 2007). Additionally, research has identified increased susceptibility to visual crowding, where letters become difficult to recognize when surrounded by other letters, which can directly impair reading speed and accuracy in individuals with dyslexia (Gori and Facoetti, 2015). While the causal role of these visual factors remains debated, empirical studies consistently demonstrate their elevated prevalence in dyslexic populations.

Consistent with these models, a significant subset of individuals with dyslexia also experiences pronounced visual stress and dysfunction. Raghuram et al. (2019) reported that vision-related symptoms including tired eyes, double vision, intermittent blurring of words, uncomfortable eyes, headaches, and words moving on the page are more frequent in children with dyslexia. While coloured lenses or filters may offer symptomatic relief, they do not constitute rehabilitative treatment for the underlying visual processing inefficiency (Suttle and Conway, 2025). This highlights a potential treatment gap where visual aspects of dyslexia are managed palliatively rather than therapeutically.

2.3. Visual-based differences and similarities of ADHD and dyslexia

Kibby et al. (2015) demonstrated that children with ADHD and dyslexia show visual processing defects that differ in the specific pathway affected, yet also overlap substantially. Children with ADHD exhibit impairments primarily in the visual-spatial processing pathway, including visual-spatial storage and the visual-spatial central executive domains of working memory (Shiels et al., 2008). In contrast, children with dyslexia show deficits in visual discrimination and sequential short-term memory (Kibby et al., 2015).

Importantly, visual attention and visuospatial working memory are not separate general cognitive functions but integral components of visual processing in both disorders (Kibby et al., 2015). Deficits in sustained visual attention, visual search efficiency, and visuospatial working memory are associated with impaired reading fluency, text comprehension, and difficulty maintaining focus on visual tasks. These higher-order visual processing impairments frequently co-occur with low-level deficits such as binocular visual dysfunction (BVD), oculomotor instability, and increased susceptibility to visual crowding (Mukhtar et al., 2024; Solé Puig et al., 2015; Stevens et al., 2012). Studies have also shown prolonged reaction times in the left visual field in both ADHD and dyslexia, possibly due to asymmetric hemispheric control of visual attention (Facoetti et al., 2001; Jones et al., 2008; Mohamed et al., 2016).

The convergence of these specific visual processing deficits—BVD, crowding, left visual field asymmetries, inefficient visual attention, and visuospatial working memory overload—provides a compelling rationale for interventions that target these overlapping neural and functional pathways (Kibby et al., 2015; Mukhtar et al., 2024; Stein, 2019). Traditional approaches to these issues—such as prism lenses, patching, or specific visual training (Aleci et al., 2015; Barton and Ranalli, 2021; Doble et al., 2010; Smaakjær et al., 2022)—can be effective but often lack the engagement needed for consistent adherence in pediatric populations. This context creates a strong opportunity for immersive technologies like VRT. VRT can deliver engaging, personalized, and intensive training of both low-level binocular/oculomotor functions and higher-order visual processing skills (visual attention, processing speed, and visuospatial working memory) in ecologically valid environments, which is particularly promising for children with ADHD and/or dyslexia.

2.4. Existing systematic reviews on VR interventions

Previous reviews have synthesized evidence on VRT, but their scopes remain segmented, failing to address the specific intersection that is the focus of this study. As summarized in Table 1, existing systematic reviews have predominantly investigated one of two avenues: (1) the efficacy of VRT for general visual dysfunctions such as amblyopia and strabismus in mixed or non-specified populations (Chan et al., 2023; Ma et al., 2022), or (2) the application of VRT for managing the core neurodevelopmental symptoms of ADHD or dyslexia, such as inattention or reading accuracy (Corrigan et al., 2023; Romero-Ayuso et al., 2021). Even reviews encompassing both disorders have focused on broad learning or cognitive outcomes rather than visual dysfunction (Lozano-Álvarez et al., 2023). Other relevant reviews have examined specific aspects such as the use of AR/VR games for ADHD (Sinha et al., 2023), VR-assisted mindfulness interventions for ADHD (Mitsea et al., 2023), and the challenges of VR in developmental disorders (AbdelMassih et al., 2024).

A critical synthesis gap remains: no systematic review has yet consolidated evidence to explicitly evaluate the role of VRT in treating visual dysfunction within the clinically complex, comorbid ADHD-dyslexia population. This gap limits the translation of VRT's potential into targeted clinical guidance for this group. Therefore, this review directly addresses this omission by posing research questions that bridge the segmented literature, aiming to evaluate the methodologies, applications, and effectiveness of VRT specifically for visual dysfunctions in children with ADHD and/or dyslexia.

3. Research objective and methodology

This systematic review aims to synthesize and critically evaluate the current evidence on the application of VRT for managing visual dysfunction in children diagnosed with ADHD and/or dyslexia. The review methodology was designed to identify, select, and analyze relevant studies in a structured and reproducible manner.

3.1. Research questions

To achieve the primary objective, this review is guided by three specific research questions (RQs), each formulated to address a distinct aspect of the literature. The rationale for each RQ is detailed in Table 2.

3.2. Search strategy

This systematic review was not preregistered in a prospective registry such as PROSPERO or Open Science Framework. The authors acknowledge this as a limitation, as preregistration helps prevent reporting bias, selective outcome reporting, and unplanned changes to analysis methods. To mitigate potential bias, the authors adhered to PRISMA guidelines, prespecified the research questions and all eligibility criteria, and fully documented the search strategy and screening process (see Fig. 1). Study selection and data extraction were performed independently by two reviewers, with any disagreements resolved by a third reviewer.

3.2.1. Search strings and source searched

A systematic search was performed across five major electronic databases: PubMed, Scopus, Web of Science, IEEE Xplore, and ScienceDirect. The search timeframe spanned from January 1, 2015, to December 31, 2025, to capture the most recent decade of technological and clinical advancements. The search strategy employed a combination of keywords and controlled vocabulary terms (e.g., MeSH in PubMed) related to three core concepts:

1. Population: "ADHD", "Attention Deficit Hyperactivity Disorder", "Dyslexia", "Reading Disorder".

Table 1
Overview of the related reviews on VR in the treatment of visual dysfunction or ADHD and dyslexia.

Article	Publication year	Topics				Areas covered				
		ADHD	Dyslexia	Virtual reality (VR)	Visual dysfunction	Relieve symptoms of ADHD using VR	Relieve symptoms of Dyslexia using VR	Treatment of visual dysfunction related to ADHD and dyslexia	Included comparison of treatments with and without VR	Drawbacks and limitations of VR
Sinha et al.	2023	Yes	No	Yes	No	Yes	No	No	No	No
Romero-Ayuso et al.	2021	Yes	No	Yes	No	Yes	No	No	Yes	No
Corrigan et al.	2023	Yes	No	Yes	No	Yes	No	No	Yes	No
Lozano-Álvarez et al.	2023	Yes	Yes	Yes	No	Yes	Yes	No	Yes	No
AbdelMassih et al.	2024	Yes	Yes	Yes	No	No	No	No	No	Yes
Mitsea et al.	2023	Yes	No	Yes	No	Yes	No	No	Yes	No
Ma et al.	2022	No	No	Yes	Yes	No	No	Yes	Yes	Yes
Chan et al.	2023	No	No	Yes	Yes	No	No	Yes	No	Yes
Our proposed paper		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Table 2
Research questions for the review.

Research question	Motivation
RQ1: What visual processing domains are targeted by VRT interventions in children with ADHD and/or dyslexia?	To identify which specific visual processing domains (e.g., visual attention, processing speed, binocular function, visuospatial working memory) VRT studies have deliberately targeted as intervention goals, rather than assuming all visual dysfunctions are present.
RQ2: What VRT methodologies and outcome measures are currently employed in treating visual dysfunction in children?	To map the technological approaches (e.g., hardware, software, gamification) and assessment tools used in existing interventions, highlighting current practices and variability.
RQ3: What is the effectiveness of VRT in treating visual dysfunctions in children with comorbid ADHD and/or dyslexia?	To directly evaluate the therapeutic efficacy of VRT for this population by synthesizing outcomes related to visual function, attention, and academic/behavioral metrics.

2. Intervention: "Virtual Reality".
3. Condition/Outcome: "Visual Dysfunction", "Binocular Dysfunction", "Convergence Insufficiency", "Visual Processing", "Amblyopia".

Boolean operators (AND, OR) were used to combine these concepts. The specific search strings for each database are detailed in Table 3. To ensure completeness, the reference lists of retrieved articles and relevant systematic reviews were manually screened to identify any further eligible studies.

3.2.2. Study selection process and eligibility criteria

All retrieved records (n = 1406) were imported into EndNote 20.4.1 for reference management. After removing 421 duplicates using automated and manual checks, 985 unique records remained for screening. The study selection involved two sequential phases:

For title or abstract screening, two reviewers independently screened the 985 records against the inclusion and exclusion criteria. Discrepancies were resolved through discussion or consultation with a third reviewer. For full-text screening, the full text of 166 potentially eligible articles was retrieved and assessed independently by two reviewers.

During full-text screening, each study was required to explicitly report at least one quantitative or qualitative outcome measure in one of the following predefined visual domains: (1) low-level visual function (e.g., best-corrected visual acuity, amblyopia, stereoacuity, near point of convergence, positive fusional vergence, accommodation, binocular dysfunction, or visual crowding); or (2) higher-order visual processing (e.g., sustained visual attention, selective visual attention, visual search efficiency, visual processing speed, visuospatial working memory, or visual perceptual skills such as discrimination and figure-ground perception). Studies were included if they reported VRT outcomes in any of these visual domains (low-level visual function or higher-order visual processing). This approach captures both primary visual interventions (with potential applicability to ADHD and dyslexia) and those targeting the visual-cognitive deficits that are commonly embedded in ADHD and dyslexia symptoms. Purely non-visual cognitive outcomes (e.g., verbal memory alone or general executive function without a clear visual processing component) were excluded.

For this review, VRT was defined as any intervention delivered through a computer-generated three-dimensional virtual environment with real-time user interaction. This included immersive VR (e.g., Oculus Quest, HTC Vive), semi-immersive VR (e.g., Kinect-based systems), and non-immersive VR (e.g., tablet-based 3D applications). Purely two-dimensional screen-based tasks without a virtual environment component were excluded. Ultimately, 20 studies satisfied all criteria and were included for data extraction and synthesis (Fig. 1).

The established eligibility criteria were crucial for ensuring a systematic examination of the selected articles. The inclusion criteria required that articles be published between 2015 and 2025, written in English, directly related to VRT, and report at least one outcome measure in the domains of visual function defined above. Conversely, articles were excluded if they fell into any of the following categories: review articles, book chapters, or conference papers; irrelevant to the use of VR; unrelated to the treatment or management of ADHD, Dyslexia, or visual dysfunction; or when the full text could not be obtained after reasonable efforts, defined as attempting retrieval via institutional subscriptions and contacting corresponding authors by email (two follow-up attempts over three weeks). For records where the full text remained unavailable, the reviewers screened the abstract; if the abstract suggested potential eligibility, the study was recorded as 'full text not retrievable' and excluded.

3.3. Quality assessment and risk of bias

To evaluate the methodological quality of the included studies, the reviewers applied validated tools appropriate to study design. Randomized controlled trials (RCTs) were assessed using the Cochrane Risk of Bias 2 (RoB 2) tool across five domains: randomization process,

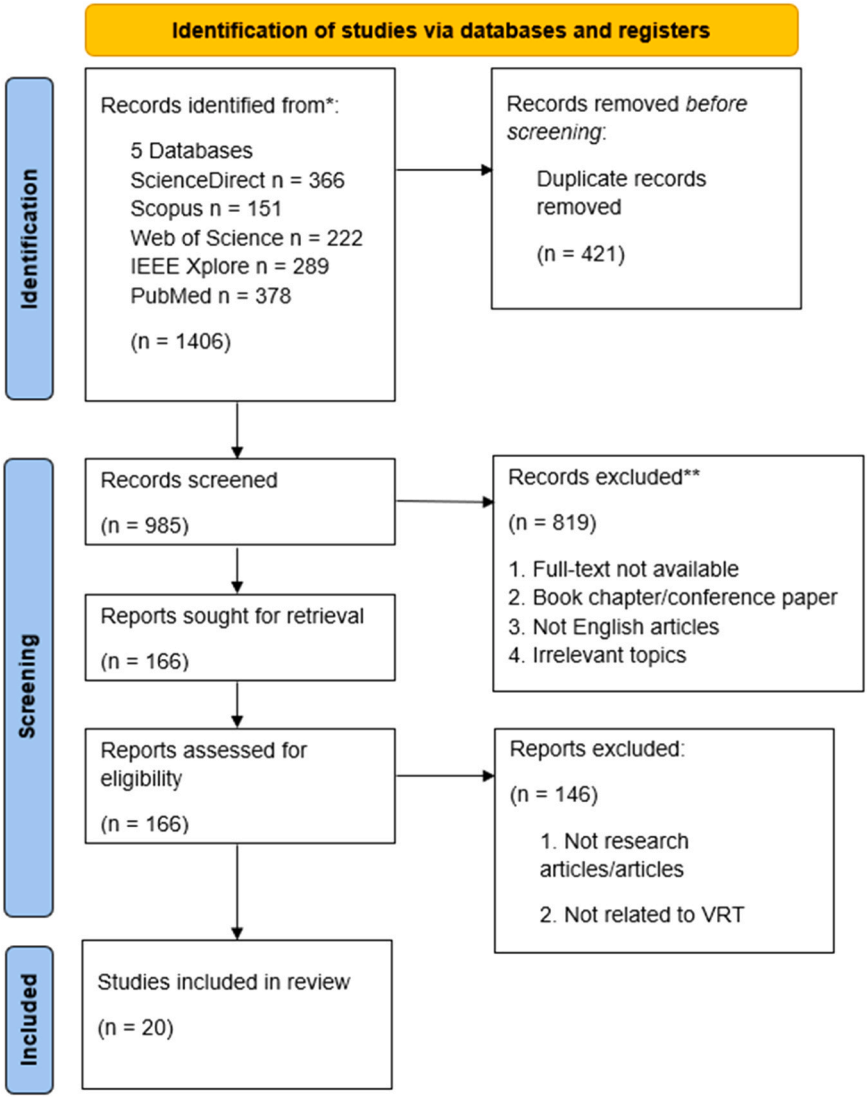


Fig. 1. *PRISMA flowchart.* *Reporting the number of records identified from each database or register searched. **Records were excluded by automation tools.

Table 3
Search strings.

Database	Results	Keyword search and other applied filters
ScienceDirect	364	("ADHD" OR "Dyslexia" OR "visual dysfunction") AND "Virtual reality" AND Year: 2015–2025. Article type: Research articles
	2	("ADHD" OR "Dyslexia") AND "visual dysfunction" AND "Virtual reality" AND Year: 2015–2025. Article type: Research articles
Scopus	151	TITLE-ABS-KEY (("ADHD" OR "Dyslexia" OR "visual dysfunction") AND "Virtual reality" AND PUBYEAR > 2015 AND (LIMIT-TO (DOCTYPE, "article")) AND (LIMIT-TO (LANGUAGE, "English"))
Web of Science	222	("ADHD" OR "Dyslexia" OR "visual dysfunction") AND "Virtual reality" AND Year: 2015–2025 AND LA = (English) AND DT = (Article)
IEEE Xplore	289	ALL = (("ADHD" OR "Dyslexia" OR "visual dysfunction") AND "Virtual reality"), Filter applied: 2015–2025, Journals
PubMed	376	(((ADHD) OR (Dyslexia)) OR (visual dysfunction)) AND (virtual reality) Filter applied: 2015–2025, English
	2	((ADHD) OR (Dyslexia)) AND (visual dysfunction) AND (virtual reality) Filter applied: 2015–2025, English

deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. Non-randomized studies, including single-group pre-post interventions, case-control studies, and two-group quasi-experimental designs, were evaluated using appropriate tools based on their design: the Newcastle-Ottawa Scale (NOS) for case-control studies (maximum 9 stars), the JBI Critical Appraisal Checklist for Quasi-Experimental Studies for single-group pre-post and two-group quasi-experimental designs, and the Mixed Methods Appraisal Tool (MMAT) for mixed-methods studies.

Two reviewers independently performed the quality assessments, with disagreements resolved by discussion or consultation with a third reviewer. Overall, the quality of evidence was moderate. Most RCTs were rated as having "some concerns" or "low risk" of bias, primarily due to challenges with blinding participants and personnel (inherent to VR interventions) and small sample sizes. Single-group intervention studies generally received lower quality ratings due to the absence of a control group, small sample sizes, and lack of blinding. Notably, Lozano-Álvarez et al. (2025) was rated as "Low" quality due to its very small sample size (n = 2), with only one VR case), absence of a control group, and lack of statistically significant findings, with the authors themselves stating that "conclusions regarding efficacy cannot be drawn due to the limited sample size". Similarly, Ou et al. (2020) (n = 3) and Pedroli et al. (2017) (n = 10) received lower ratings for comparable reasons. No studies were excluded based on quality, but findings from lower-quality studies were

interpreted with greater caution and were not prioritized in the synthesis of intervention effectiveness. The full quality assessment of all 20 included studies is presented in Appendix A (Table A1). A summary of the quality ratings is provided in Table 4.

4. Result

This section provides a comprehensive overview of VRT in relation to ADHD, dyslexia, and visual dysfunction. It also addresses the research questions outlined in the previous section. The summary of the included studies is shown in Table 5.

4.1. Research trends and characteristics

A total of 20 studies met the inclusion criteria, providing a contemporary snapshot of VRT research intersecting visual dysfunction, ADHD, and dyslexia. The publication trend (Fig. 2) reveals a marked increase in research activity, with 17 studies (85%) published since 2021, peaking in 2024 (6 studies). Notably, only 1 study from 2025 was included in the final analysis. This is not indicative of a decline in research interest but is primarily attributable to the inherent time lag in the academic publication and indexing cycle. Research conducted or submitted in 2025 is unlikely to have completed peer review, been published in final form, and been fully indexed in the databases searched by the time. Most 2025 records retrieved during the search were excluded upon full-text screening, typically because they were conference proceedings, study protocols, or articles that did not meet the specific population-intervention-outcome criteria. Therefore, the concentration of included studies from 2021 to 2024 accurately reflects the most recent completed and accessible body of empirical work, underscoring the growing recognition of VRT's potential in these clinical domains.

To highlight the research landscape, a visual depiction of the distribution of disorders examined across all included studies is presented in Fig. 3. The review analysis showed that most studies are directed toward ADHD (45%), followed by attention to dyslexia (35%), and amblyopia (15%). The results emphasize the current research priorities and point out potential gaps in exploring comorbid conditions.

The technological landscape was diverse, employing a range of VR hardware (Fig. 4). PC/Console-Powered VR headsets (e.g., HTC Vive) were the most prevalent (n = 7), followed by mobile-powered VR solutions, including smartphone-based and tablet-based systems (n = 5) and standalone VR headsets (e.g., Meta Quest, Oculus Quest 2) (n = 4). This diversity highlights the parallel use of accessible, consumer-grade VR and higher-fidelity, immersive systems for therapeutic applications.

4.2. RQ1: What visual processing domains are targeted by virtual reality therapy (VRT) interventions in children with ADHD and/or dyslexia?

The 20 included studies recruited children and adolescents with ADHD, dyslexia, or specific learning disabilities in whom VRT interventions targeted visual processing deficits as part of their clinical profile. This synthesis reveals that VRT interventions were deliberately designed to address the overlapping visual dysfunction profiles commonly observed in ADHD and dyslexia. Table 6 summarizes the most frequently reported or implied visual processing domains that were explicitly targeted (or strongly implied as targets) across the included studies.

The results show that VRT interventions in the included studies primarily targeted a range of overlapping visual processing domains in children and adolescents with ADHD and/or dyslexia. The most consistently addressed domain was visual attention and vigilance, with multiple studies focusing on difficulties sustaining visual focus, impaired visual search efficiency, and reduced accuracy/speed on visual tasks (Alobidi et al., 2024; Di Giusto et al., 2023; Fang et al., 2019; Flores-Gallegos et al., 2022; Kim et al., 2024; Skalski et al., 2021).

Visual processing speed emerged as another key target, with interventions addressing slower and more erratic processing of visual information, as well as inefficient visual information-seeking strategies (Cunha et al., 2023; Kim et al., 2024; Pedroli et al., 2017). Challenges in oculomotor and binocular control were also frequently targeted, including inefficient visual scanning, poor fixation control, and oculomotor instability (Flores-Gallegos et al., 2022; Kim et al., 2024).

Visuospatial working memory and visual perceptual challenges, such as visual discrimination, figure-ground perception, and visual crowding, were targeted in several studies, often as part of broader efforts to reduce cognitive overload during academic tasks (Köse et al., 2022; Kim et al., 2024; Lozano-Álvarez et al., 2025).

A key finding is the substantial overlap in the visual processing domains targeted across both ADHD and dyslexia populations. Interventions frequently addressed shared impairments in sustained visual attention, visual search, processing speed, oculomotor control, and visual perception. These converging deficits extend beyond the core diagnostic criteria of each disorder and contribute significantly to functional academic difficulties, particularly in reading fluency, text comprehension, and overall school performance (Maresca et al., 2022).

In summary, VR-based visual rehabilitation interventions predominantly target multiple integrated visual processing domains—spanning low-level, such as processing speed, binocular/oculomotor control, and higher-order, such as attentional vigilance, visuospatial working memory functions in children with ADHD and/or dyslexia. By focusing on these overlapping visual impairments, VRT offers a promising, transdiagnostic approach to reducing visual-related academic disability.

4.3. RQ2: What VRT methodologies and outcome measures are currently employed in treating visual dysfunction in children?

VRT methodologies and outcome measures employed in the included studies reflect the multifaceted nature of visual processing deficits in ADHD and dyslexia. Across the reviewed papers, VRT applications can be usefully categorized along a continuum rather than as entirely separate approaches.

For primary visual dysfunctions (e.g., amblyopia and convergence insufficiency), VRT predominantly utilizes dichoptic stimulation and vergence/accommodation training to target low-level binocular and oculomotor deficits. In contrast, interventions addressing ADHD- and dyslexia-related visual processing deficits employ gamified, adaptive, immersive platforms to train higher-order visual skills—such as sustained visual attention, visual search efficiency, perceptual discrimination, visuospatial working memory, and resistance to visual crowding—that are specifically impaired in these neurodevelopmental disorders (see Table 5 for domain-specific manifestations).

This distinction is meaningful: although all VR experiences have a visual component, the studies in this review deliberately target the precise visual processing pathways shown to be dysfunctional in ADHD and dyslexia, rather than generic cognitive training. The immersive nature of VRT allows simultaneous engagement of low-level visual

Table 4
Summary of the quality ratings.

Quality Rating	Number of Studies	Examples
Good	1	Fang et al. (2019)
Low risk	4	Köse et al. (2022); Li et al. (2022); Meqdad et al. (2024); Skalski et al. (2021)
Some concerns	10	Alobidi et al. (2024); Cunha et al. (2023); De Luca et al. (2024); Elhusseiny et al. (2021); Kim et al. (2024); Maresca et al. (2022); Maresca et al. (2024); Rajavi et al. (2021); Schena et al. (2023)
Moderate	4	Cuber et al. (2024); Di Giusto et al. (2023); Flores-Gallegos et al. (2022);
Low	3	Ou et al. (2020); Lozano-Álvarez et al. (2025); Pedroli et al. (2017)

Table 5
Summary of the included studies.

Reference	Type of disorder	No. of participant	Mean age (years ±S.D.)	Study design	Objective	Main outcome	Device used*
Alobidi et al. (2024)	ADHD	Control group: 10; Experiment group: 10	N/A	Randomized Controlled Trial (RCT)	To investigate the impact of VR interventions on improving the learning outcomes, with a specific emphasis on addressing some symptoms of ADHD and academic functioning.	Significant post-test improvements in learning outcomes (animal vocabulary) in VR group vs. control.	Oculus Quest 2
Cuber et al. (2024)	ADHD	Single group: 27	21.04 (±1.60)	Mixed-methods study	To investigate whether VR can improve concentration, motivation, and effort during academic tasks for university students with ADHD.	Significant gains in concentration/task efficiency during VR; reduced distractions and improved sustained attention in real-world tasks.	Varjo XR-3 VR headset
Cunha et al. (2023)	ADHD	Control group: 13; Experiment group: 13	20.17 (± 0.94) for control group; 21.69 (± 0.75) for VR group	Randomized Controlled Trial (RCT)	To evaluate the effectiveness of a virtual-reality-based intervention in processing speed and working memory in students with ADHD symptomatology.	Significant improvement in processing speed in the VR group, with no significant change in the control group; minor improvement in visuospatial working memory (Spatial Location test) in the VR group only.	Meta Quest VR headset
De Luca et al. (2024)	ADHD	Control group: 30; Experiment group: 30	N/A	Randomized Controlled Trial (RCT)	To evaluate the effectiveness of the BRAVO platform (adaptive serious games in virtual/extended reality) for treating ADHD in children.	Significant improvements in selective/sustained attention, motor inhibition (BIA frog test), and grammatical comprehension vs. control	HTC Vive + Kinect + Emotiv Epoc EEG + Empatica E4 wristband
Di Giusto et al. (2023)	Dyslexia	Single group: 24	8.9	Single-group pre-post intervention	To examine whether a VR-based rehabilitation program could improve Executive Functions (EFs) in children with SLD.	Significant visual attention improvements from baseline (T0) to post-intervention (T1), maintained at follow-up (T2).	Semi-immersive setup: VITAMIN platform*
Elhusseiny et al. (2021)	Amblyopia	Full-treatment group: 11; Sham-crossover: 9	N/A	Randomized Controlled Trial (RCT)	To evaluate whether a VR-based binocular therapy prototype can improve visual acuity and stereoacuity in treatment-resistant patients with amblyopia.	No significant visual acuity gains; significant stereoacuity improvements at 8/16 weeks vs. baseline (clinically minor).	Apple iPhone 6 Plus + Zeiss VR One Plus headset
Fang et al. (2019)	ADHD	ADHD group: 77; Control group: 63	8.34 (± 1.41) for ADHD group; 8.17 (± 1.54) for control group	Case-control study	To explore the feasibility and availability of VR for evaluating symptoms of ADHD in school-aged children with learning problems.	ADHD group showed lower VR accuracy (87.66% vs. 91.08%), fewer correct items (36.82 vs. 38.92), longer times (237.6 s vs. 211.33 s) vs. controls; strong correlations with IVA-CPT/CBCL attention scores.	VRMC system + high-end VR headset
Flores-Gallegos et al. (2022)	Dyslexia	Control group: 5; Experiment group: 6	7.9	Two-group quasi-experimental	To explore the effects of a VR training program on reading, visual attention, motor balance, and self-perception in children with RLD.	Significant visual attention improvements (ENI-2) and reduced omission errors (TOVA); strong correlation with reading speed; large effect sizes in reading accuracy/speed/comprehension.	Oculus Rift VR headset
Kim et al. (2024)	ADHD	Control group: 12; Experiment group: 12	10.75	Between-subjects RCT	To investigate whether a novel VR training system using breathing input can simultaneously improve visual attention and pulmonary function in children with ADHD.	Significant visual attention improvements (ATA): reduced omission/commission errors; overall ATA gains vs. controls.	Samsung Gear VR + Galaxy S20 smartphone

(*continued on next page*)

Table 5 (*continued*)

Reference	Type of disorder	No. of participant	Mean age (years ±S.D.)	Study design	Objective	Main outcome	Device used*
Köse et al. (2022)	Dyslexia	Control group: 69; Study group: 69	8.21 (± 1.02) for Therapist Guided Game-based Intervention Program group; 8.10 (± 0.95) for the control group	Single-blind RCT	To investigate the effects of a therapist-guided (actively supervised) versus a self-oriented (non-supervised) VR game-based intervention program on the visual perception skills of children with SLD.	Significant visual perception improvements in both groups post-8 weeks; therapist-guided (TGGIP) superior (d=0.93) vs. unsupervised (GIP; d=0.67).	Non-immersive setup: iPad Mini tablet (7.9-inch)
Li et al. (2022)	convergence insufficiency or accommodative dysfunction	VR group: 33; OBVAT group: 30	23.63 (±1.75) for Virtual reality-based vision Therapy; 23.53 (±2.00) for OBVAT	Randomized Controlled Trial (RCT)	To compare the effectiveness of VR-based vision therapy versus traditional OBVAT for improving binocular vision in young adults with CI or AD.	Significant binocular vision gains (NPC, PFV, accommodative amplitude/facility) in both groups post-12 weeks; VR equivalent to OBVAT for vergence, slightly less for accommodation.	Unspecified HMD with dual screens
Lozano-Álvarez et al. (2025)	Dyslexia	VR case: 1; AR case: 1	13	Single-group pre-post case study	To explore the implementation and educational impact of a VR/AR-based assistive technology developed to support learners with dyslexia, assessing cognitive changes via NEPSY-II and user experience via TAM.	Positive trends in attention, inhibition, narrative memory, and phonological processing; high motivation and satisfaction; no statistically significant results; conclusions regarding efficacy cannot be drawn due to limited sample size.	Oculus Quest 2
Maresca et al. (2022)	Dyslexia	Control group: 14; Experiment group: 14	10.3 (±2.0)	Randomized Controlled Trial (RCT)	To evaluate the efficacy of a virtual reality rehabilitation system (VRRS) on cognitive and reading/writing abilities in children with dyslexia.	Significant gains in word-reading/homophone writing, perceptual reasoning (PRI), and processing speed (PSI) on WISC-IV vs. conventional treatment.	VRRS (Virtual Reality Rehabilitation System)
Maresca et al. (2024)	Dyslexia	Control group: 12; Experiment group: 12	10.2 (±1.9)	Randomized Controlled Trial (RCT)	To evaluate the one-year effects of VR neurorehabilitation training (VRNT) on reading skills in children with dyslexia.	Significant long-term gains in non-word reading/reading speed (maintained at 12 months); greater verbal comprehension/global IQ improvements vs. controls.	VRRS (Virtual Reality Rehabilitation System)
Meqdad et al. (2024)	Amblyopia	Patching group: 42; VR group: 44	13.31 (± 6.47)	Randomized Controlled Trial (RCT)	To compare the effectiveness of dichoptic stimulation using VR versus conventional patching for treating moderate and severe amblyopia.	Significant BCVA improvements in both groups (10 weeks: VR +0.89 lines, patching +1.38; 20 weeks: VR +1.32, patching +1.0); no inter-group difference, but VR superior for adults/severe cases.	Oculus Rift OC CV1 VR HMD
Ou et al. (2020)	ADHD	Single group: 3	10	Single-group pre-post intervention	To develop and test immersive VR rehabilitation games to improve attention, cognitive ability, abstract reasoning, and complex information processing in children with ADHD.	Improvement in attention (ATESC) and parent-reported reductions in inattention, hyperactivity/impulsivity, and oppositional defiance (SNAP-IV).	HTC VIVE VR system
Pedroli et al. (2017)	Dyslexia	Single group: 10	10.6 ± 1.4	Single-group pre-post design	To propose and assess VR training for attention to improve the reading skills in a sample of dyslexic children.	Significant improvement in attentional skills (Posner task: cued/unequal targets); trend toward faster low-frequency word reading, but no immediate reading gains.	Semi-immersive setup: Laptop + Microsoft Kinect + audio system
Rajavi et al. (2021)	Amblyopia	Patching group: 25; VR group: 25	7.1 (±1.80)	Randomized Controlled Trial (RCT)	To compare the visual outcome of occlusion therapy with VR game	Significant BCVA improvements in both groups; VR superior by	Smartphone-based VR headset

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Table 5 (*continued*)

Reference	Type of disorder	No. of participant	Mean age (years ±S.D.)	Study design	Objective	Main outcome	Device used*
					playing as a new therapy for children with amblyopia.	letter count ($p = 0.002$), similar by logMAR.	
Schena et al. (2023)	ADHD	Control group: 30; Experiment group: 30	8	Randomized Controlled Trial (RCT)	To evaluate the effectiveness of the VR serious game "IAmHero" on ADHD symptoms and executive functions.	Significant improvements in sustained visual attention, selective auditory attention, learning problems; enhanced executive functions (planning/ problem-solving).	Oculus Quest 2 VR headset
Skalski et al. (2021)	ADHD	Standard HEG BFB group: 30; VR HEG BFB with distractors group: 28; BFB without distractors group: 29	12	Three-armed RCT	To compare the effects of HEG biofeedback delivered in VR vs. standard 2D environments on attention functions in children with ADHD.	Greater improvements in vigilance, visual search, and divided attention with VR HEG biofeedback vs. 2D; no difference with/without distractors.	HTC Vive Cosmos Elite HMD

* Device with medical device approval

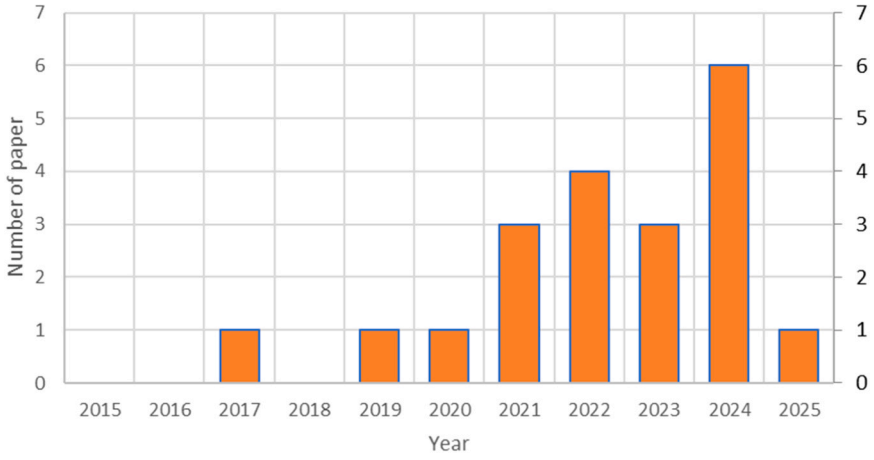


Fig. 2. Publication trends in VRT research.

functions and higher-order visual-cognitive integration in ecologically valid environments, which differentiates these approaches from non-immersive interventions reviewed elsewhere.

4.3.1. *VRT methodologies for primary visual dysfunction*

For amblyopia, VRT predominantly utilizes dichoptic stimulation to reduce interocular suppression. Studies by Meqdad et al. (2024) and Rajavi et al. (2021) employed platforms such as Vivid Vision on Oculus Rift or smartphone-based headsets, presenting separate images to each eye with contrast modulation to strengthen the amblyopic eye. Protocols varied in intensity and duration: Meqdad et al. (2024) implemented sessions of 2 h weekly for 10 weeks, whereas Rajavi et al. (2021) used a more intensive regimen of 1 h per day, 5 days per week, for 4 weeks. Elhusseiny et al. (2021) targeted treatment-resistant amblyopia in older children and adults using a Zeiss VR One Plus headset with an iPhone, delivering binocular therapy through flicker and contrast adjustment in flexible 1-hour daily sessions over 8 weeks.

For convergence insufficiency (CI) and accommodative dysfunction, Li et al. (2022) implemented vergence and accommodation training using a head-mounted display (HMD) with dual screens for dichoptic stimulation. Exercises required coordinated eye movements and focusing at varying distances over a 12-week period, with task difficulty adapting to performance.

4.3.2. *VRT methodologies for visual processing deficits in neurodevelopmental disorders*

In the context of ADHD, dyslexia, and specific learning disorders, VRT focuses on enhancing visual attention, processing speed, and perceptual skills through gamified, adaptive tasks.

For attention and vigilance training, Kim et al. (2024) used a Samsung Gear VR system integrated with respiratory biofeedback, where breathing patterns controlled gameplay to train sustained visual attention over 8 training sessions within four weeks. Similarly, Flores-Gallegos et al. (2022) employed Oculus Rift games like *Beat Saber* and *GrafoTami* to improve visual attention and motor coordination in children with reading disabilities through a 30-session training program.

For integrated visual attention and motor inhibition training in children with SLD, Di Giusto et al. (2023) implemented a semi-immersive VR program using the VITAMIN platform (a class I medical device) and performed the game-based VR training over 6 weeks. These games were designed to simultaneously train visual selective attention, inhibitory control, visuospatial working memory, and goal-directed motor planning in an adaptive, game-based format.

For executive function and processing speed, De Luca et al. (2024) developed the BRAVO platform, an integrated multimodal system combining HTC Vive (immersive VR), EEG (Emotiv Epoc 14-channel) for neurofeedback, and Kinect sensors for motion tracking. This platform

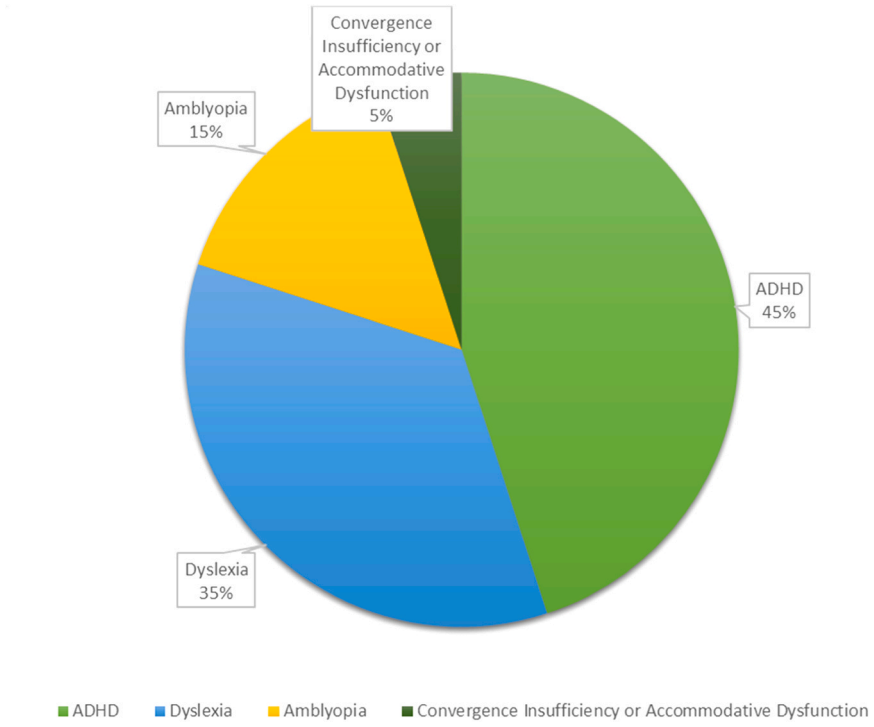


Fig. 3. Proportion of studies by disorder type.

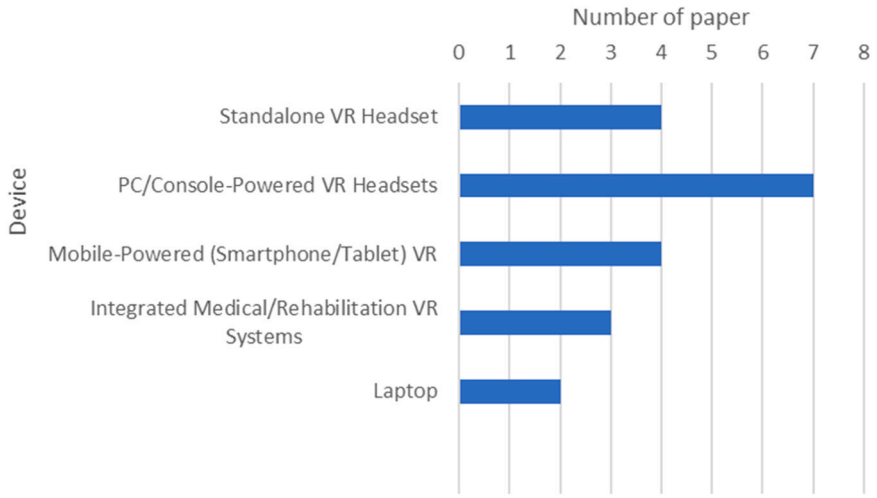


Fig. 4. VR Device usage across studies.

delivered adaptive serious games over a 6-month intervention, targeting selective attention, sustained visual attention, and processing speed through three distinct game-based modules. Cunha et al. (2023) conducted a randomized controlled pilot study using Meta Quest (standalone VR headset) to evaluate the impact of a VR-based cognitive training intervention on visual processing speed (PS) and visuospatial working memory (WM) in young adults with ADHD. The intervention employed the Enhance VR platform, comprising six gamified tasks delivered in 10 sessions (30 min each, twice weekly over five weeks), with task difficulty adapting in real time based on performance.

For visual perception and discrimination, Köse et al. (2022) examined the role of therapist guidance within a VRT framework for children with Specific Learning Disabilities (SLD). Their study used a commercially available, non-immersive VR game *Brawl Stars* delivered via an iPad, comparing a Therapist-Guided Game-Based Intervention Program (TGGIP) with a self-guided Game-Based Intervention Program (GIP). Both groups engaged in 45-minute sessions twice weekly for 8 weeks.

For ecological validity in assessment, Fang et al. (2019) utilized a high-fidelity VR system (VRMC) with a high-end VR headset to create immersive classroom-like environments for evaluating visual attention deficits. The VR assessment consisted of three gamified modules targeting visuospatial attention and motor control, completed in approximately 20 min, and demonstrated strong correlations with traditional tests such as the IVA-CPT.

A common thread across these methodologies is the use of consumer-grade VR hardware (standalone, PC-tethered, or smartphone-based) and the integration of gamification, adaptive difficulty, and real-time feedback to maintain engagement and personalize therapy.

4.3.3. Outcome measures

The studies reviewed utilized a variety of standardized and study-specific outcome measures to evaluate the effectiveness of VRT across

Table 6
Visual processing domains targeted by VRT interventions in children with ADHD and/or dyslexia.

Dysfunction Domain	Manifestations in ADHD	Manifestations in Dyslexia	Shared/Overlapping Features	Functional Academic Impact
Visual Attention and Vigilance	• Inattention and difficulty sustaining focus on visual tasks (Alobidi et al., 2024; Di Giusto et al., 2023; Skalski et al., 2021). • Impaired vigilance and visual search (Kim et al., 2024; Skalski et al., 2021). • Lower accuracy and longer completion times on ecologically valid visual tasks (Fang et al., 2019).	• Impairs grapheme-phoneme coding during reading (Flores-Gallegos et al., 2022).	• Sustained visual attention deficits • Visual search difficulties	• Difficulty with text comprehension (Kim et al., 2024). • Increased academic impairments (Alobidi et al., 2024; Cuber et al., 2024; De Luca et al., 2024; Kim et al., 2024; Ou et al., 2020; Schena et al., 2023).
Visual Processing and Speed	• Erratic/slower visual processing speed (Cunha et al., 2023). • Inefficient visual information-seeking strategies (Kim et al., 2024).	• Slow processing speed for rapid visual stimuli (Pedroli et al., 2017).	• Slower and more variable visual processing speed	• Reduced reading fluency and comprehension (Maresca et al., 2024, 2022; Pedroli et al., 2017).
Oculomotor and Binocular Control	• Less efficient visual scanning/ fixation control (Kim et al., 2024).	• Associated with greater saccadic amplitude during reading (Flores-Gallegos et al., 2022).	• Binocular visual dysfunction (BVD) and oculomotor instability • Difficulty maintaining visual fixation on a task • Inefficient visual scanning patterns	-
Visuospatial Working Memory	• Pronounced deficits in visual-spatial working memory (Kim et al., 2024).	• Reading inefficiency overloads visual-spatial resources (Lozano-Álvarez et al., 2025).	• Visual-spatial working memory deficits	-
Visual Perceptual Challenges	• Impaired visual information-seeking strategies (Kim et al., 2024).	• Deficits in visual discrimination, figure-ground perception, visual memory (Köse et al., 2022).	• Increased visual crowding • Broad visual perception deficits	-

*Note: Academic impairments refers to reduced performance in school-related visual tasks (e.g., reading textbooks, following visual instructions, completing worksheets) that arise directly from the visual processing deficits listed above. These are not general cognitive or behavioral symptoms but specific functional consequences of inefficient visual attention, slow processing speed, poor oculomotor control, and visuospatial working memory overload.

different visual dysfunctions.

In treatments targeting primary visual conditions, measures often included best-corrected visual acuity (BCVA) and stereoacuity for amblyopia, and near point of convergence (NPC), positive fusional vergence (PFV), and the Convergence Insufficiency Symptom Survey (CISS) for convergence insufficiency and accommodative dysfunction (Li et al., 2022; Meqdad et al., 2024; Rajavi et al., 2021).

For interventions addressing visual processing deficits in ADHD, dyslexia, and related learning disabilities, outcome measures were more diverse, encompassing both cognitive-behavioral and performance-based metrics. These included standardized tools such as the Italian Battery for ADHD (BIA), which assesses auditory and visual attention, working memory, and inhibition; the Visual Motor Integration (VMI) test for visual-perceptual and motor coordination; the Language Battery (BVL) for verbal fluency and working memory; and the Motor-Free Visual Perception Test-3 (MVPT-3) for visual perception independent of motor output (De Luca et al., 2024; Di Giusto et al., 2023; Köse et al., 2022). Studies also employed neuropsychological batteries tailored to specific visual-cognitive domains; for instance, Di Giusto et al. (2023) used the Visual Attention and Design Fluency subtests of NEPSY-II to quantify gains in visual scanning speed, accuracy, and visuospatial planning following VR training in children with SLD. Additionally, studies frequently incorporated neuropsychological tests (e.g., Southwestern Assessment of Processing Speed, WAIS-III subtests), behavioral rating scales (e.g., IVA-CPT, CBCL, CPRS), and in-game performance metrics (e.g., task completion time, error rates, accuracy) to capture changes in visual attention, processing speed, and executive function (Cunha et al., 2023; Fang et al., 2019; Flores-Gallegos et al., 2022; Kim et al., 2024). This multi-method approach allows for a comprehensive evaluation of VRT's impact on both low-level visual functions and higher-order cognitive processes.

4.4. RQ3: What is the effectiveness of VRT in the treatment of visual dysfunctions related to ADHD and dyslexia?

Although few studies directly examined children with comorbid ADHD and dyslexia, the included trials targeting ADHD or dyslexia consistently evaluated VRT effects on shared visual processing outcomes (low-level and higher-order). This provides indirect yet valuable evidence for the potential utility of VRT in the comorbid population.

Research on VRT has shown the effectiveness of VRT in addressing a range of visual dysfunctions associated with ADHD, dyslexia, and specific learning disorders, with benefits observed across multiple functional domains. Summary of the review on the effectiveness of VRT in treating visual dysfunctions related to ADHD and dyslexia is shown in Table 7. The systematic review reveals a range of functional domains, including best-corrected visual acuity (BCVA), stereoacuity, visual attention, visual perception, reading, motor coordination, binocular functions and processing time.

For amblyopia, VRT has demonstrated efficacy in improving BCVA and stereoacuity, often comparable to traditional patching therapy. For instance, Meqdad et al. (2024) reported BCVA gains of 0.89 lines after 10 weeks of VRT, increasing to 1.32 lines after 20 weeks, with no statistically significant difference compared to patching. Rajavi et al. (2021) similarly found significant BCVA improvements in both VRT and patching groups, though VRT showed greater gains in letter-count assessments ($P = 0.002$). Notably, Elhusseiny et al. (2021) observed no significant BCVA change after 8 weeks of VRT but documented meaningful improvements in stereoacuity, underscoring VRT's potential to enhance binocular function even when monocular acuity remains stable.

For convergence insufficiency and accommodative dysfunction, VRT has proven to be a viable alternative to office-based vision therapy (OBVAT). Li et al. (2022) reported significant improvements in binocular functions—including NPC, PFV, and CISS scores—in both VRT and OBVAT groups after 12 weeks, though VRT showed slightly lesser gains

Table 7
Summary of the review on the effectiveness of VRT in treating ADHD and dyslexia.

Study	BCVA	Stereoacuity	Visual Attention	Visual Perception	Reading	Motor Coordination	Binocular Functions	Processing Time
Meqdad et al. (2024)	✓							
Rajavi et al. (2021)	✓				✓			
Elhusseiny et al. (2021)		✓						
Li et al. (2022)							✓	
De Luca et al. (2024)			✓			✓		
Kim et al. (2024)			✓					
Cunha et al. (2023)			✓					✓
Flores-Gallegos et al. (2022)			✓		✓	✓		
Köse et al. (2022)			✓	✓				
Di Giusto et al. (2023)			✓					
Fang et al. (2019)			✓					

in monocular accommodative facility. These findings suggest that VRT can effectively address vergence and accommodation deficits, supporting its use as a complementary or alternative treatment modality.

In the context of neurodevelopmental disorders, VRT has shown positive effects on visual attention, processing speed, executive functions, and academic-related skills. For children with ADHD, interventions such as the BRAVO platform (De Luca et al., 2024) led to significant time × group interaction effects in selective attention ($F=5.69$, $p = 0.03$), though the interaction for visual-motor integration was not significant. Pulmonary VR training (Kim et al., 2024) resulted in significant reductions in omission errors ($t = 4.799$, $p < 0.001$), commission errors ($t = 3.566$, $p < 0.01$) in visual attention, and improvements in the visual Advanced Test of Attention (ATA) ($t = 5.106$, $p < 0.001$). Similarly, Cunha et al. (2023) found that a gamified VR intervention significantly enhanced processing speed (time × group interaction: $p = 0.004$, $\eta p^2=0.309$), with SWAPS scores increasing from 40.77 ± 6.94–49.46 ± 6.96 ($p < 0.001$) in the VR group, while visuospatial working memory also improved within the VR group ($p = 0.034$). For dyslexia and reading learning disabilities (RLD), VRT has been associated with gains in visual attention, reading speed, and motor coordination. Flores-Gallegos et al. (2022) reported a strong positive correlation between visual attention and reading speed following VR training ($r = 0.992$, $p = 0.001$), with visual attention improvement showing a significant between-group difference ($p = 0.043$). Moreover, stand-alone game-based VRT without therapist guidance has been shown to improve visual perception skills in children with specific learning disabilities (Köse et al., 2022). The game-based group demonstrated significant improvements in total MVPT-3 scores (pre: 19.05 ± 3.27 to post: 20.26 ± 3.72, $p = 0.0001$). Additionally, Di Giusto et al. (2023) found that a 6-week VR program significantly improved visual attention ($p = 0.006$) in children with SLD, with gains in attention and inhibition largely maintained at 6-month follow-up.

Beyond therapeutic application, VR also shows promise as an assessment tool for visual attention deficits in ADHD. Fang et al. (2019) demonstrated that VR-based performance metrics strongly correlated with traditional neuropsychological tests and parent-reported behavioral scales. VR accuracy significantly correlated with visual attention scores from the IVA-CPT ($r = 0.192$, $p = 0.025$), and total time correlated with visual ($r = -0.222$, $p = 0.009$), highlighting VR's potential for objective evaluation and progress monitoring.

In summary, VRT demonstrates multifaceted effectiveness across visual dysfunction subtypes, with consistent evidence supporting its utility in improving visual acuity, binocular function, attention, processing speed, and perceptual skills in children with ADHD, dyslexia, and related conditions.

5. Discussion

This systematic review synthesizes evidence on virtual reality therapy (VRT) for the full spectrum of visual processing deficits in children with ADHD and/or dyslexia. By defining visual dysfunction to include both low-level visual functions (e.g., binocular control, accommodation, and stereopsis) and higher-order visual-cognitive domains (e.g., visual attention, visual processing speed, and visuospatial working memory), the review reveals VRT's distinctive advantage: immersive environments enable targeted, simultaneous training of the precise visual pathways that are dysfunctional in these disorders, unlike traditional therapies that often address symptoms in isolation.

In addressing RQ1, the synthesized evidence demonstrates that VRT interventions systematically target a shared profile of visual processing impairments common to ADHD and dyslexia. These impairments extend beyond the core diagnostic criteria of each disorder and significantly contribute to academic difficulties. Across studies, interventions most strongly targeted visual attention and vigilance—improving sustained visual focus, visual search efficiency, and attentional control—while also addressing visual processing speed, oculomotor/binocular control (e.g., fixation stability and scanning), and, in some cases, visuospatial working memory and broader visual perceptual deficits.

Although most included studies examined ADHD or dyslexia separately rather than in comorbid samples, the substantial overlap in targeted visual processing domains identified in RQ1 and Table 5 suggests shared neurobiological underpinnings that VRT is particularly well-suited to address in a transdiagnostic manner. Dyslexia is consistently associated with inefficiencies in the magnocellular-dorsal visual pathway, which supports rapid temporal processing, motion detection, and precise oculomotor control—functions essential for stable fixation and fluent reading— though this remains debated (Gori and Facoetti, 2015; Stein, 2019). In ADHD, similar behavioral deficits in oculomotor control and processing speed have been documented (Bellato et al., 2023), though the magnocellular theory originates from dyslexia research. In contrast to behavioral and cognitive overlaps, a large-scale meta-analysis by McGrath and Stoodley (2019) found that the gray matter correlates of dyslexia and ADHD are largely distinct. Only a single region of subtle convergence was identified in the right caudate, and this was observed only under a lenient statistical threshold, potentially implicating shared mechanisms in procedural learning rather than widespread attentional networks.

VRT offers a distinct advantage here because its immersive and ecologically valid environments allow simultaneous training of these shared low-level (e.g., binocular stability and oculomotor control) and higher-order (e.g., sustained visual attention, visual processing speed, and visuospatial working memory) deficits within a single adaptive platform. By providing gamified tasks with real-time feedback and dynamic difficulty adjustment, VRT may promote neuroplastic changes along the dorsal stream and attentional networks, thereby reducing the visual-cognitive load that underlies academic impairments common to both disorders. This transdiagnostic mechanism distinguishes VRT from traditional unimodal interventions and supports its potential as an integrated rehabilitation approach, particularly for children with comorbid ADHD and dyslexia. Future studies should explicitly test these shared visual pathways in comorbid populations to evaluate additive therapeutic benefits.

Regarding RQ2, the review maps a diverse yet coherent methodological landscape. The distinction noted in RQ2—dichoptic and vergence training for primary visual conditions versus gamified visual-perceptual training for neurodevelopmental disorders—highlights VRT's versatility rather than a conceptual weakness. Primary visual studies provide important mechanistic proof-of-concept (e.g., improving binocular control), while ADHD- and dyslexia-focused studies demonstrate real-world transfer to academically relevant visual processing outcomes such as reading fluency and sustained visual attention. This integration is novel; prior meta-analyses of VR for ADHD have focused primarily on broad cognitive outcomes without specifically emphasizing visual processing, and reviews on visual dysfunction have rarely addressed neurodevelopmental comorbidity.

For RQ3, the analysis demonstrates that VRT is effective in improving specific visual dysfunctions associated with these conditions. For amblyopia, studies such as Meqdad et al. (2024) and Rajavi et al. (2021) report significant gains in best-corrected visual acuity (BCVA) and stereoacuity, with efficacy comparable to traditional patching. For convergence insufficiency, VRT has proven a viable alternative to office-based vision therapy, improving binocular functions like near point of convergence and positive fusional vergence (Li et al., 2022). In the realm of neurodevelopmental disorders, VRT shows positive effects on core domains such as visual attention, processing speed, and visual perception in children with ADHD and dyslexia (Cunha et al., 2023; Flores-Gallegos et al., 2022; Köse et al., 2022). This is further supported by Di Giusto et al. (2023), whose VR intervention led to significant and sustained improvements in visual attention and inhibitory control in children with SLD, underscoring VRT's potential to target shared visual-cognitive deficits across learning disorders.

One of the key advantages of VRT lies in its capacity to address both low-level and higher-order visual processing deficits within the same intervention platform. The engaging, adaptive, and ecologically valid characteristics of immersive VR make it especially suitable for children who often struggle with traditional therapy adherence. The gamified elements of VR tools not only captivate users but also encourage consistent participation in therapy sessions, as indicated by Rajavi et al. (2021). This is particularly beneficial for children and young adults, who may find traditional methods less motivating. The ability to use VR at home also facilitates earlier detection of visual changes and promotes ongoing engagement, which is crucial for effective treatment outcomes.

However, it is important to note that not all studies included in this review measured both low-level and higher-order visual outcomes simultaneously. Consequently, the degree to which VRT interventions exert distinct or synergistic effects across these two levels of visual processing remains unclear. Future research should explicitly incorporate outcome measures from both domains within the same study design to empirically validate the integrated framework proposed here.

While several studies reported limited improvements in visual acuity, notable gains in stereoacuity and other binocular functions highlight VRT's capacity to address complex visual processing issues. For instance, Elhusseiny et al. (2021) highlight VRT's capacity to address higher-order visual integration and depth perception. This suggests that VRT may operate through mechanisms that enhance visual integration and depth perception, which are critical for overall visual function.

However, the review for RQ3 also reveals a significant synthesis gap: very few studies specifically investigated VRT for visual dysfunction within the comorbid ADHD-dyslexia population. Most existing research examines each disorder or general visual conditions in isolation. This limits definitive conclusions regarding VRT's tailored efficacy for this complex, high-need group and represents a critical avenue for future clinical research.

Moreover, the versatility of VRT, as demonstrated across different studies, indicates its applicability for various visual dysfunctions, from amblyopia to convergence insufficiency. The tailored approaches outlined in the studies, ranging from dichoptic stimulation to vergence training, and from gamified attention training (e.g., Di Giusto et al., 2023; Kim et al., 2024) to integrated neurofeedback systems (De Luca et al., 2024), emphasize the technology's flexibility to meet heterogeneous patient needs. This adaptability is essential for addressing the individual variability seen in conditions like ADHD and dyslexia, where symptoms can manifest differently across individuals.

Looking forward, the integration of AI with VRT promises to move beyond a universal therapeutic approach. AI can enable real-time performance monitoring and dynamic adjustment of difficulty, creating a truly personalized treatment protocol. This approach not only optimizes the therapeutic dosage for each individual but also increases patient engagement by maintaining an optimal challenge level. The combination of personalized, data-driven therapy with the inherently engaging nature of VRT could significantly improve the management of visual dysfunction associated with ADHD and dyslexia. The evidence from these studies suggests that VRT is a valuable tool in the diagnosis and treatment of various visual conditions, and further exploration of its long-term benefits and broader clinical applications is needed.

6. Limitations and future works

This systematic review is subject to several key limitations that reflect the current state of the literature and outline important directions for future research. A primary limitation is the scarcity of studies directly addressing the comorbid ADHD-dyslexia population in the context of visual dysfunction, a significant gap identified while answering RQ3. The existing literature remains fragmented and imbalanced, with most research focusing on either ADHD or dyslexia in isolation and few studies targeting the shared visual dysfunction profile outlined in RQ1 within a truly comorbid sample. This restricts definitive conclusions about VRT's tailored efficacy for this clinically complex group. Beyond the scarcity of comorbid samples, the overall quality of the available evidence also presents a challenge. While the findings suggest that VRT may improve visual attention and processing speed in children with ADHD and dyslexia, the overall low to moderate quality of the evidence—characterized by small sample sizes, lack of blinding, and methodological heterogeneity—limits the strength of conclusions regarding its definitive effectiveness. Larger, well-designed RCTs are needed to confirm these preliminary observations.

A further limitation is the conceptual breadth of visual processing, which necessarily includes both low-level (e.g., binocular and oculomotor functions) and higher-order domains (e.g., visual attention, processing speed, and visuospatial working memory). This heterogeneity, while firmly grounded in the neurodevelopmental literature, complicates direct comparisons and meta-analytic synthesis across studies.

Additionally, methodological heterogeneity presents a substantial challenge, as considerable variability exists across studies in VRT protocols, hardware platforms, and outcome measures—diversity that was mapped in RQ2 but complicates synthesis. Frequent small sample sizes and a general lack of long-term follow-up in the reviewed studies further limit robust conclusions regarding optimal treatment parameters and the sustainability of therapeutic benefits. While every effort was made to obtain full texts through interlibrary loans and author contact, some potentially relevant studies could not be retrieved and were excluded, introducing possible availability bias.

In light of these limitations, future research should prioritize several interconnected avenues. First, studies must directly address the identified synthesis gap by recruiting children with comorbid ADHD and dyslexia and designing VRT interventions explicitly tailored to their common visual dysfunctions. Second, to strengthen the evidence base, researchers should pursue larger, more methodologically rigorous studies employing standardized protocols, core outcome sets, and longitudinal designs to evaluate lasting effects. Finally, future work should explore the integration of artificial intelligence for adaptive, personalized therapy and investigate hybrid models that combine VRT with other evidence-based interventions. Such advances would move the field toward more comprehensive and effective treatment strategies for this

population.

7. Conclusion

This systematic review synthesizes studies on the use of VRT for treating visual dysfunctions in children with ADHD and dyslexia. The findings confirm that VRT interventions target a common profile of visual impairments affecting domains such as visual attention and vigilance, visual processing speed, oculomotor and binocular control, visuospatial working memory, and visual perceptual challenges in children with ADHD and/or dyslexia, establishing a clear therapeutic target. The review further maps the diverse yet converging methodologies of VRT, which leverage engaging, adaptive, and often gamified platforms to address these dysfunctions. Importantly, the analysis demonstrates that VRT is an effective intervention for improving specific visual outcomes—including visual acuity, stereoacuity, and visual attention—with efficacy comparable or superior to traditional therapies, particularly in terms of user engagement and compliance.

The engaging, flexible, and increasingly personalized nature of VRT positions it as a highly promising tool for pediatric neurovisual rehabilitation. However, its full potential for the comorbid ADHD-dyslexia population remains underexplored due to a lack of targeted research, and the evidence base is currently constrained by methodological variability and short-term study designs. Therefore, while VRT represents a significant advancement in treatment modality, future research must prioritize well-controlled, longitudinal studies that directly address visual dysfunction in children with comorbid ADHD and dyslexia, employ standardized methods, and explore integrated, technology-enhanced therapeutic models to translate this promise into optimized clinical practice.

Funding

This project is supported by the Research Centre for Assistive Technology (RCATech) of the Hong Kong Polytechnic University (Project Ref.: CE0E), and the Research Talent Hub for SPC projects (RTH-SPC) of HKSAR, China (Project Ref.: IsP/205/24 and IcP/063/25), for the research, authorship and/or publication of this article.

Declaration of Competing Interest

The authors report no commercial or proprietary interest in any product or concept discussed in this article.

Appendix

Table A1
Quality assessment of included studies

Study	Study Design	Quality Assessment Tool	Overall Quality Rating	Main Concerns / Limitations
Alobidi et al. (2024)	RCT	RoB 2	Some concerns	Lack of blinding, small sample size
Cuber et al. (2024)	Mixed-methods	MMAT	Moderate	No control group, self-report bias, mostly inattentive subtype
Cunha et al. (2023)	RCT	RoB 2	Some concerns	Small sample, short-term follow-up
De Luca et al. (2024)	RCT	RoB 2	Some concerns	Lack of blinding
Di Giusto et al. (2023)	Single-group pre-post intervention	JBI	Moderate	No control group, small sample, lack of blinding
Elhusseiny et al. (2021)	RCT	RoB 2	Some concerns	Treatment-resistant cases, small sample
Fang et al. (2019)	Case-control study	NOS	Good (7/9)	Selection bias possible
Flores-Gallegos et al. (2022)	Two-group quasi-experimental	JBI	Moderate	Lack of blinding, small sample size
Kim et al. (2024)	Between-subjects RCT	RoB 2	Some concerns	Blinding difficult
Köse et al. (2022)	Single-blind RCT	RoB 2	Low risk	Reliance on commercial game (Brawl Stars) – potential for updates/ changes, no long-term follow-up
Li et al. (2022)	RCT	RoB 2	Low risk	Short-term follow-up
Lozano-Álvarez et al. (2025)	Single-group pre-post intervention	JBI	Low	Very small sample (n = 2), no control group, no statistical significance
Maresca et al. (2022)	RCT	RoB 2	Some concerns	Small sample, blinding poorly reported
Maresca et al. (2024)	RCT	RoB 2	Some concerns	Long-term follow-up attrition
Meqdad et al. (2024)	RCT	RoB 2	Low risk	Blinding challenges
Ou et al. (2020)	Single-group pre-post intervention	JBI	Low	Very small sample (n = 3), no control group; lack of blinding
Pedroli et al. (2017)	Single-group pre-post design	JBI	Low to Moderate	No control group, small sample, no follow-up
Rajavi et al., 2021	RCT	RoB 2	Some concerns	Blinding issues
Schena et al. (2023)	RCT	RoB 2	Some concerns	Lack of blinding, small sample size
Skalski et al. (2021)	Three-armed RCT	RoB 2	Low risk	Lack of blinding

Data availability

The data and materials are available from the corresponding author on request.

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